

Hospital Visit & Resource Management using SQL

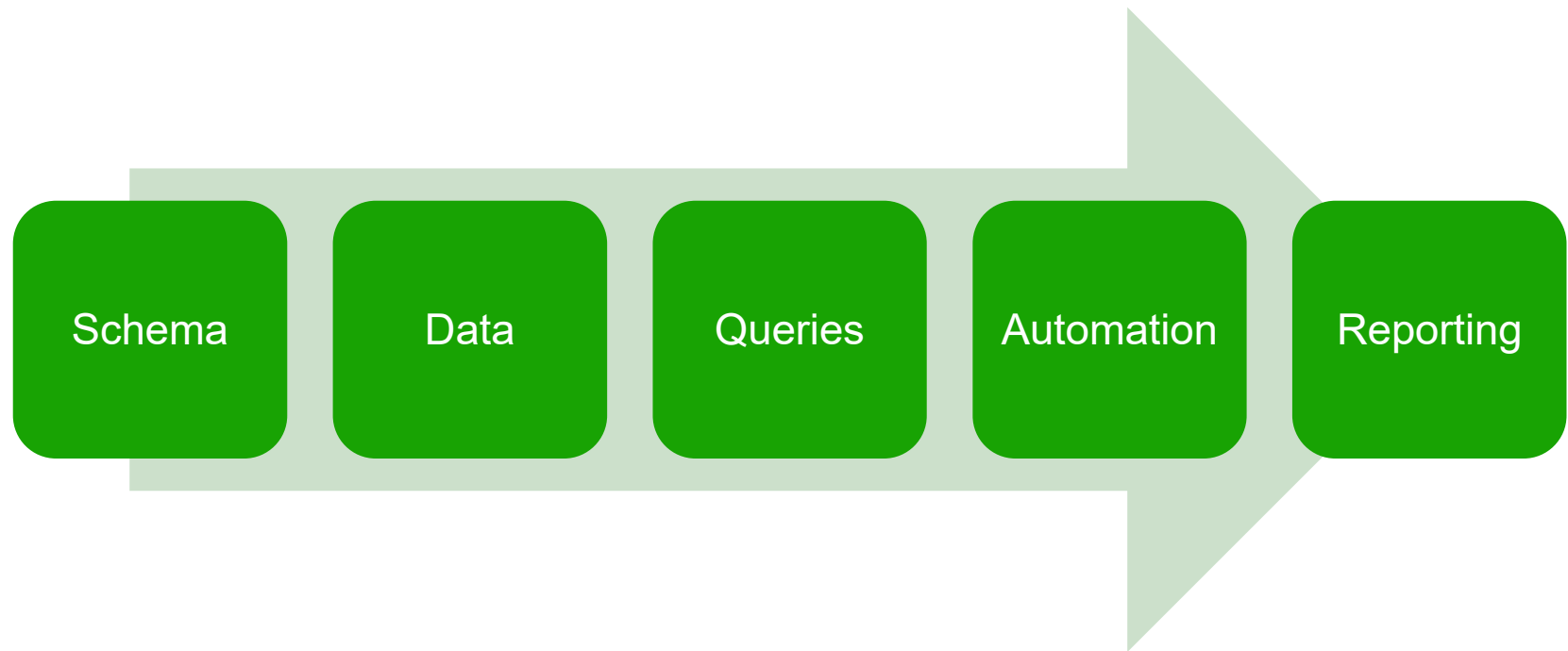
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Introduction

- Hospitals generate large volumes of patient and operational data daily.
- A structured database is required to manage visits, services, and billing efficiently.
- SQL enables accurate data storage, automation, and reporting.
- This project simulates a real-world hospital management backend system.

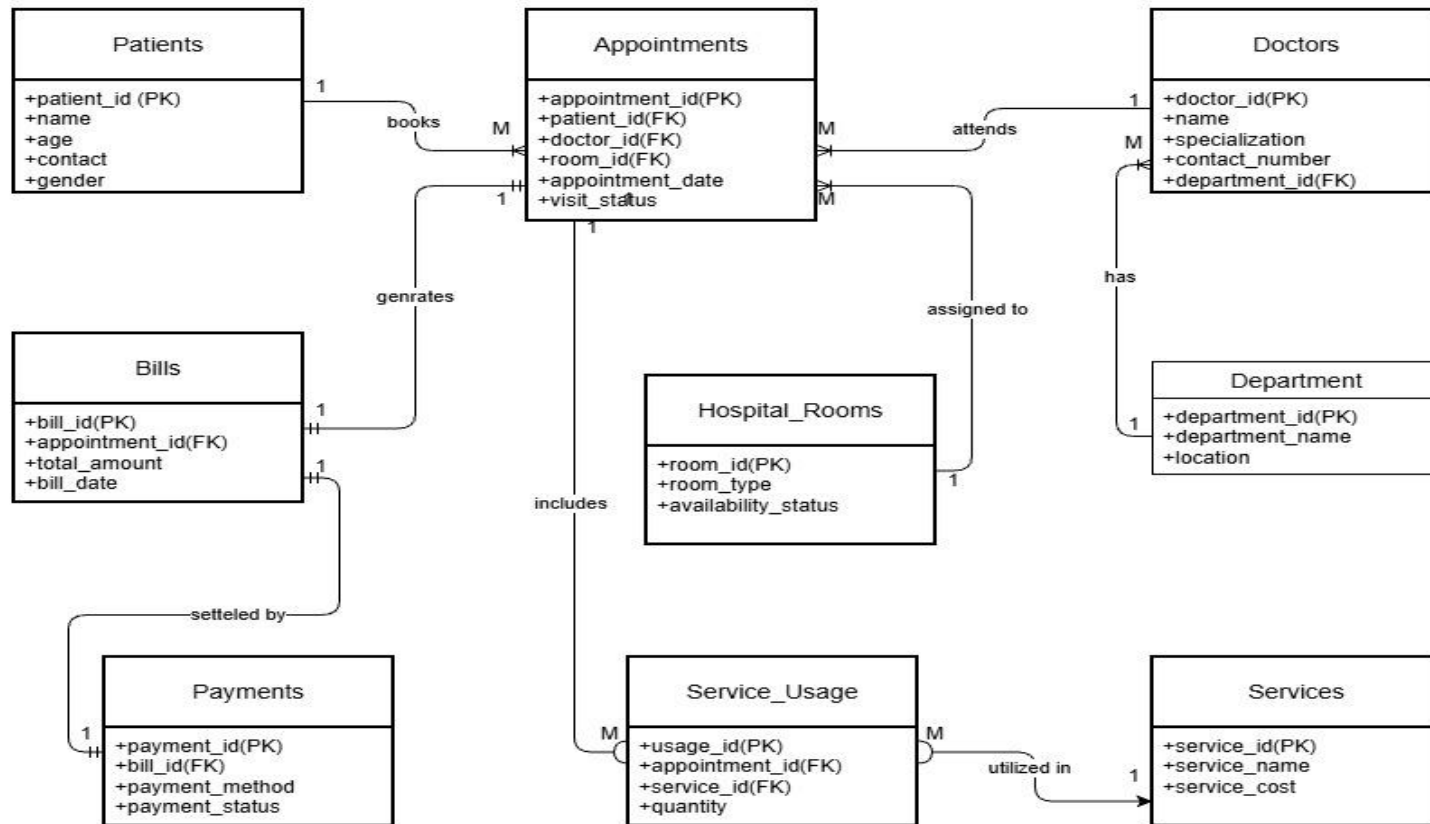
Flow



Dataset Overview

- System Type: Hospital Management Database.
- Number of Tables: 9
- Key Entities: Patients, Doctors, Services, Bills
- Database Engine: MySQL

Entity Relationship Diagram.



Project Objectives

- O1: Managing Hospital Data Efficiently.
- O2: Connecting Hospital Information Across Departments.
- O3: Automating Hospital Workflows.
- O4: Handling Flexible and Modern Data Formats.
- O5: Ensuring Reliable and Secure Operations.

Objective 1: Managing Hospital Data Efficiently

Task 1 & 2:- Creating Database Tables & Adding Hospitals Records

	patient_id	name	age	contact
▶	1	Rahul Sharma	35	9000000000
	2	Priya Patel	28	9123456780
	3	Amit Verma	42	9988776655
	4	Neha Gupta	31	9090909090
*	NULL	NULL	NULL	NULL

Patients Table

	department_id	department_name
▶	1	Cardiology
	4	General Medicine
	2	Neurology
	3	Orthopedics
*	NULL	NULL

Departments Table

	doctor_id	name	specialization	department_id
▶	1	Dr. Mehta	Cardiologist	1
	2	Dr. Singh	Neurologist	2
	3	Dr. Rao	Orthopedic Surgeon	3
	4	Dr. Khan	Physician	4
*	NULL	NULL	NULL	NULL

Doctors Table

	room_id	room_number	room_type	status
▶	1	R101	General	Available
	2	R102	ICU	Occupied
	3	R103	Private	Occupied
	4	R104	General	Maintenance
*	NULL	NULL	NULL	NULL

*Hospital_Room
Table*

Objective 1: Managing Hospital Data Efficiently

	bill_id	appointment_id	total_amount	bill_date
▶	1	1	1700.00	2026-02-05 19:06:31
	2	2	2100.00	2026-02-05 19:06:31
	3	3	5000.00	2026-02-05 19:06:31
	4	4	1300.00	2026-02-05 19:06:31
	5	1	1700.00	2026-02-06 21:10:44
*	NULL	NULL	NULL	NULL

Bill Table

	usage_id	appointment_id	service_id	quantity
▶	1	1	1	1
	2	1	2	1
	3	2	1	1
	4	2	3	2
	5	3	4	1
	6	4	1	1
	7	4	3	1
*	NULL	NULL	NULL	NULL

Service_Usage Table

	payment_id	bill_id	amount	payment_date	method
▶	1	1	1700.00	2026-02-05 19:09:55	Card
	2	2	1000.00	2026-02-05 19:09:55	Cash
	3	2	1100.00	2026-02-05 19:09:55	Card
	4	3	5000.00	2026-02-05 19:09:55	Online
	5	4	1300.00	2026-02-05 19:09:55	Insurance
	6	1	200.00	2026-02-07 11:07:27	Cash
*	NULL	NULL	NULL	NULL	NULL

Payments Table

	service_id	service_name	cost
▶	1	Consultation	500.00
	2	X-Ray	1200.00
	3	Blood Test	800.00
	4	MRI Scan	5000.00
*	NULL	NULL	NULL

Service Table

Objective 1: Managing Hospital Data Efficiently

Task 3:- Retrieve, Update, Delete Data Form Tables

	patient_id	name	age	contact
▶	1	Rahul Sharma	35	9876543211
	2	Priya Patel	28	9123456780
	3	Amit Verma	42	9988776655
	4	Neha Gupta	31	9090909090
	6	Test Patient	26	9999995555
★	NULL	NULL	NULL	NULL

	patient_id	name	age	contact
▶	1	Rahul Sharma	35	9876543211
	2	Priya Patel	28	9123456780
	3	Amit Verma	42	9988776655
	4	Neha Gupta	31	9090909090
★	NULL	NULL	NULL	NULL

	name	age
▶	Rahul Sharma	35
	Amit Verma	42
	Neha Gupta	31

- This objective demonstrates how hospital data can be safely created, modified, removed, and retrieved.
- It shows full control over the patient database and ensures that records remain accurate and up-to-date.

Objective 2: Connecting Hospital Information Across Departments

Task 1: Join patients with doctors

	patient	doctor	visit_date
▶	Rahul Sharma	Dr. Mehta	2026-02-05 19:04:10
	Rahul Sharma	Dr. Singh	2026-02-06 20:43:59
	Priya Patel	Dr. Singh	2026-02-05 19:04:10
	Priya Patel	Dr. Mehta	2026-02-07 10:47:11
	Amit Verma	Dr. Rao	2026-02-05 19:04:10
	Neha Gupta	Dr. Khan	2026-02-05 19:04:10

Task 2: Revenue Generated per Patient

	patient	total_bill
	Rahul Sharma	3400.00
	Priya Patel	2100.00
	Amit Verma	5000.00
	Neha Gupta	1300.00

Task 3: Identify Most used Hospital Services

	service_name	total_usage
▶	Consultation	3
	Blood Test	3
	X-Ray	1
	MRI Scan	1

Objective 2: Connecting Hospital Information Across Departments

Task 4: Show patients even if they have no bills (LEFT JOIN)

	name	total_amount
▶	Rahul Sharma	1700.00
	Rahul Sharma	1700.00
	Rahul Sharma	NULL
	Priya Patel	2100.00
	Priya Patel	NULL
	Amit Verma	5000.00
	Neha Gupta	1300.00

Task 5: Show all doctors including unused ones

	name	appointment_id
▶	Dr. Mehta	1
	Dr. Mehta	6
	Dr. Singh	2
	Dr. Singh	5
	Dr. Rao	3
	Dr. Khan	4

- This objective shows how hospital information from different departments can be connected using relational joins.
- It enables reporting, analysis, and cross-table insights that support better hospital decision-making.

Objective 3: Automating Hospital Workflows

Task 1: Creating stored procedure to add a new appointment

	appointment_id	patient_id	doctor_id	room_id	visit_date
▶	1	1	1	1	2026-02-05 19:04:10
	2	2	2	2	2026-02-05 19:04:10
	3	3	3	3	2026-02-05 19:04:10
	4	4	4	1	2026-02-05 19:04:10
	5	1	2	3	2026-02-06 20:43:59
	6	2	1	3	2026-02-07 10:47:11
	9	2	3	1	2026-02-09 12:25:03
•	NULL	NULL	NULL	NULL	NULL

Task 2: stored procedure to add Payment

	payment_id	bill_id	amount	payment_date	method
▶	1	1	1700.00	2026-02-05 19:09:55	Card
	2	2	1000.00	2026-02-05 19:09:55	Cash
	3	2	1100.00	2026-02-05 19:09:55	Card
	4	3	5000.00	2026-02-05 19:09:55	Online
	5	4	1300.00	2026-02-05 19:09:55	Insurance
	6	1	200.00	2026-02-07 11:07:27	Cash
	7	5	500.00	2026-02-09 12:27:57	Cash
•	NULL	NULL	NULL	NULL	NULL

Task 3: Creating stored procedure to calculate bill

	bill_id	appointment_id	total_amount	bill_date
▶	1	1	1700.00	2026-02-05 19:06:31
	2	2	2100.00	2026-02-05 19:06:31
	3	3	5000.00	2026-02-05 19:06:31
	4	4	1300.00	2026-02-05 19:06:31
	5	1	1700.00	2026-02-06 21:10:44
	6	1	1700.00	2026-02-09 12:24:33
	7	3	5000.00	2026-02-09 12:25:50
•	NULL	NULL	NULL	NULL

- This objective demonstrates how automation reduces manual work in hospital systems.
- Stored procedures ensure consistent workflows and minimize human errors during billing and appointment management.

Objective 4: Handling Flexible & Modern Data Formats

Task 1: Alter Appointments table & add JSON 'medical_notes' Column

	appointment_id	medical_notes
▶	1	{"symptoms": "fever and cough", "diagnosis": "viral infection", "doctor_notes": "rest for 5 days"}
★	NULL	NULL

Task 2: Extracting JSON data, values

	appointment_id	diagnosis
▶	1	viral infection

Task 3: Extracting JSON data, values

	appointment_id
▶	1

- This objective shows how modern databases handle flexible and semi-structured data using JSON.
- It allows hospitals to store dynamic medical information without redesigning the schema.

Objective 5: Creating Ensuring Reliable & Secure Operations

Task 1: Trigger to auto update room status after adding in Appointments

	room_id	room_number	room_type	status
▶	1	R101	General	Occupied
	2	R102	ICU	Occupied
	3	R103	Private	Occupied
	4	R104	General	Maintenance
*	NULL	NULL	NULL	NULL



	room_id	room_number	room_type	status
▶	1	R101	General	Occupied
	2	R102	ICU	Occupied
	3	R103	Private	Occupied
	4	R104	General	Occupied
*	NULL	NULL	NULL	NULL

Objective 5: Creating Ensuring Reliable & Secure Operations

Task 2: Implementing Transaction in SQL

payment_id	bill_id	amount	payment_date	method
3	2	1100.00	2026-02-05 19:09:55	Card
4	3	5000.00	2026-02-05 19:09:55	Online
5	4	1300.00	2026-02-05 19:09:55	Insurance
6	1	200.00	2026-02-07 11:07:27	Cash
7	5	500.00	2026-02-09 12:27:57	Cash
8	6	800.00	2026-02-09 12:35:27	Cash
9	1	9999.00	2026-02-09 12:36:18	Error
NULL	NULL	NULL	NULL	NULL

RollBack

payment_id	bill_id	amount	payment_date	method
2	2	1000.00	2026-02-05 19:09:55	Cash
3	2	1100.00	2026-02-05 19:09:55	Card
4	3	5000.00	2026-02-05 19:09:55	Online
5	4	1300.00	2026-02-05 19:09:55	Insurance
6	1	200.00	2026-02-07 11:07:27	Cash
7	5	500.00	2026-02-09 12:27:57	Cash
8	6	800.00	2026-02-09 12:35:27	Cash
NULL	NULL	NULL	NULL	NULL

Objective 5: Creating Ensuring Reliable & Secure Operations

Task 3: Create a Hospital View

	patient	doctor	total_amount	visit_date
▶	Rahul Sharma	Dr. Mehta	1700.00	2026-02-05 19:04:10
	Rahul Sharma	Dr. Mehta	1700.00	2026-02-05 19:04:10
	Rahul Sharma	Dr. Mehta	1700.00	2026-02-05 19:04:10
	Priya Patel	Dr. Singh	2100.00	2026-02-05 19:04:10
	Amit Verma	Dr. Rao	5000.00	2026-02-05 19:04:10
	Amit Verma	Dr. Rao	5000.00	2026-02-05 19:04:10
	Neha Gupta	Dr. Khan	1300.00	2026-02-05 19:04:10

- This objective ensures database reliability and safety.
- Triggers, transactions, and constraints protect hospital data from errors and maintain system consistency during critical operations.

Challenges Faced

- Managing Relational Dependencies.
- Designing Normalized Schema.
- JSON integration complexity.

Conclusion

- SQL implementation improved data integrity and enabled automation.
- The System demonstrates real-world hospital database design.